Transfer Learning for Cross-Market Commodity Return Forecasting

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1 Introduction

1.1 Motivation & Background

Some of the more complex models seen in Section 2.1.1 suffer from needing of a lot of data. For some more niche markets with shorter histories and more constrained datasets this raises a problem. The most straightforward and perhaps naive solution is to simply train models on data-rich markets and implement them on data-sparse ones. However, it has been shown that traditional models fail when being transferred cross-market. When trying to transfer a convolutional neural network (CNN) model trained on US commodity futures to predict returns of the Chinese market, Hao et al. [1] find that this model fails to systematically outperform models trained on data specific to the Chinese market. These results suggest that there are little significant return-predictive patterns from US commodity futures which generalize across different markets. Hao et al. [1] give two explanations on why employing models cross-market fails in commodity markets. insert reasons Furthermore, Nguyen and Walther [2] have also found results to be heterogeneous across different commodity futures as no single model fit all commodities. It seems that traditional models lack cross-market transferability across both different countries and commodities.

1.2 The Research Gap

1.3 Research Questions & Hypotheses

1.4 Contributions to the Literature

1.5 Thesis Outline

2 Literature Review

This section will review the existing literature on the topic and work out the research gap this thesis aims to explore. Section 2.1 goes over existing approaches used in commodity forecasting and presents previous and remaining challenges. Section 2.4 explores the field of transfer learning as a potential solution for some of these remaining challenges. Section 2.5 shows where the existing literature leaves a gap that this thesis aims to fill, as the intersection of transfer learning, commodity markets and data-sparse emerging economies.

2.1 Commodity Forecasting

Before evaluating whether transfer learning improves commodity forecasting in data-sparse markets, it is necessary to establish what classical, single-market methods already achieve when data is not scarce. This benchmark matters for two reasons: classical models supply the target-only baselines agains which the proposed framework is compared, see Section 4.5, and their known limitations, even under the most favourable data conditions, motivate the case for knowledge transfer in the first place.

2.1.1 Classical Approaches

A large body of empirical literature deals with regressions of commodity series on macroeconomic and financial predictors and evaluates the in-sample and out-of-sample forecasting performance. For instance, Gargano and Timmermann [3] study the predictive power over different forecast horizons, economic states, and types of commodities. Using Reuters/Jefferies-CRB commodity indexes over 1947-2010, they find that predictability from variables such as industrial production growth, commodity-currency exchange rates and global economic activity indices is present but varies highly between different commodities. While it is strongest for industrials, metals and the broad commodity index, and weaker for fats/oils, foods and livestock. Furthermore, they find forecasting performance to be more pronounced during recessions. Previously, Chen, Rogoff, and Rossi [4] showed that the exchange rates of only five commodity exporters (Australia, Canada, New Zealand, South Africa and Chile) have solid forecasting power over a global aggregate commodity price index made up by more than forty products traded on various exchanges and published by the IMF. Further studies find more limited results for the predictability of commodity spot indices through commodity exchange rates and other macroeconomic variables [5]. Guidolin and Pedio [6] examine whether commodity-specific factors (basis, hedging pressure and momentum) improve forecasting performance. The idea is to test whether they include information not already captured by common macroeconomic variables. While including commodity-specific variables does not improve the forecasting perfomance, their importance seems to depend on whether statistical or economic loss functions are used. This apparent disconnect between statistical forecast accuracy of returns and the economic value created when included in a portfolio choice problem is also present in other studies [7, 8].

2.1.2 Machine Learning & Deep Learning in Commodity Markets

Over the past two decades commodity forecasting has shifted from classical statistical approaches toward machine learning and deep learning techniques. Traditional models such as autoregressive integrated moving average (ARIMA), generalized autoregressive conditional heteroskedasticity (GARCH) and exponential smoothing rely on assumptions of stationarity and linearity that frequently fail for commodity price series characterized by nonlinearity, volatility clustering and structural breaks [9, 10]. Machine learning methods like support vector regression (SVR) or extreme gradient boosting (XGBoost) provide greater flexibility in modeling multidimensional relationships without imposing strict distributional assumptions [11]. Deep learning architectures, particularly multilayer perceptron (MLP), recurrent neural network (RNN), long short-term memory (LSTM), gated recurrent unit (GRU), echo state model (ESN) and bidirectional long short-term memory (BiLSTM) have further advanced the field by capturing long-term temporal dependencies and complex nonlinear patterns [11, 10]. Next to individual approaches, hybrid models that combine decomposition techniques, ensemble methods and multiple neural architectures have emerged as a dominant trend. Fang et al. [12] use a combination of three models, support vector machine (SVM), neural network (NN) and ARIMA, to produce short-term forecasts and find that it outperforms the individual ones. Ray et al. [13] report similar results with a comparable ensemble model using a monthly forecasting period.

The literature spans multiple commodity sectors, with agricultural prices receiving a lot of attention due to their volatility and implications for food security [11, 14]. Energy commodities, especially crude oil, constitute a second cluster, driven by their geopolitical importance and market liquidity [9, 15, 16]. Studies like those from Varshini, Kayal, and Maiti [17] or Jin and Xu [18] cover forecasting of precious and industrial metals as another category, with both general economic and financial importance.

2.2 Data Scarcity in Emerging Economies

2.3 Cross-Market Commodity Dynamics

2.4 Transfer Learning: Foundations & Applications

This shortcoming naturally invites the exploration of methodologies, which aim to built predictors, that allow for differences cross-market. The goal is to leverage data-rich markets, while still being able to include market-specific characteristics. This approach can be categorised as a transfer learning technique, which is a subset of machine learning, that allows for the assumption of different domains, tasks and distributions between the training and test set [19].

2.4.1 What is Transfer Learning?

2.4.2 Transfer Learning in Finance

2.4.3 Domain Adaptation Techniques

Mashetty et al. [20] explain approach This approach proved to outperform both a traditional machine learning model and a deep learning model, which were trained only on target data, in all three chosen metrics, namely mean absolute error (MAE), root mean squared error (RMSE) and directional accuracy.

2.5 Research Gap & Positioning

3 Data

This section will describe the empirical landscape of commodity markets and available data. Subsection 3.1 addresses the reasons for the specific choice of commodity classes examined. Subsection 3.2 and 3.3 describe the data used for the source and target market respectively.

3.1 Commodity Classification & Selection

3.2 Source Market: Global Benchmark Exchanges

3.3 Target Markets: Data-Sparse Economies

Table 1
Possible Target Data Sources/Series

Source	Freq.	Commodities	Dates
DCE (Dalian Commodity Exchange)	Daily	No. 1 Soybean, No. 2 Soybean, Soybean Meal, Soybean Oil, RBD Palm Olein, Corn, Corn Starch, Pol RG Rice, Egg, Live Hogs, Fiberboard, Blockboard, Log, Coking Coal, Coke, Iron Ore, LLDPE, LLDPE MAF, PVC, PVC MAF, Polypropylene, Polypropylene MAF, Ethylene Glycol, Ethenylbenzene, LPG, Benzene	2006-2026
Zhengdong Commodity Exchange	Daily	Apple, Cotton, Chinese Jujube, Cotton Yarn, Flat Glass, Japonica Rice, Late Rice, Methanol, Rapeseed Oil, Polyester Staple Fiber, Wheat PM, Early Rice, Rapeseed Meal, Rapeseed, Soda Ash, Ferrosilicon, Manganese Silicon, White Sugar, PTA, Urea, Wheat WH, Thermal Coal, Peanut Kernel, Paraxylene, Sodium Hydroxide, Polyethylene Terephthalate Resin for Bottles, Propylene	2015-2026
Multi Commodity Exchange (MCX) India	Daily	Aluminium, Cardamom, Copper, Cotton, Cotton Oil, Crude Oil, ELECDMBL, Gold, Goldguinea, Goldm, Goldpetal, Goldten, Kapas, Lead, Leadmini, MCXBULLDEX, MCXMETLDEX, Metha Oil, Natural Gas Mini, Natural Gas, Nickel, Silver, Silver 100, Silver M, Silver MIC, Steel Rebar, Zinc, Zinc Mini	2003-2026

3.4 Feature Engineering

3.5 Descriptive Statistics & Stylised Facts

3.6 Train/Validation/Test Split

4 Methodology

This section will build the transfer learning framework used to test whether forecasting performance can be improved compared to known approaches. Subsection 4.1 presents the three-stage pipeline, which mirrors the methodology of Mashetty et al. [20], and explains how the stages connect and what information flows between them. Subsections 4.2, 4.3 and 4.4 each explore one of the three stages: source pre-training, domain adaptation as well as fine-tuning and target prediction.

4.1 Framework Overview

This section introduces the research design aimed at answering the research questions introduced in Section 1.3. The goal throughout this thesis is to forecast the h-period-ahead log return of a commodity traded in a data-scarce target market,

$$r_{t+h} = \ln P_{t+h} - \ln P_t \tag{1}$$

where P_t denotes the commodity price at time t and h is the forecast horizon. Returns are used as the prediction target rather than price levels because commodity price series are typically non-stationary, whereas returns provide a more suitable basis for forecasting while retaining a clear economic interpretation. The choice of h depends on the length and characteristics of the available target-market sample and is justified in Section 3.6.

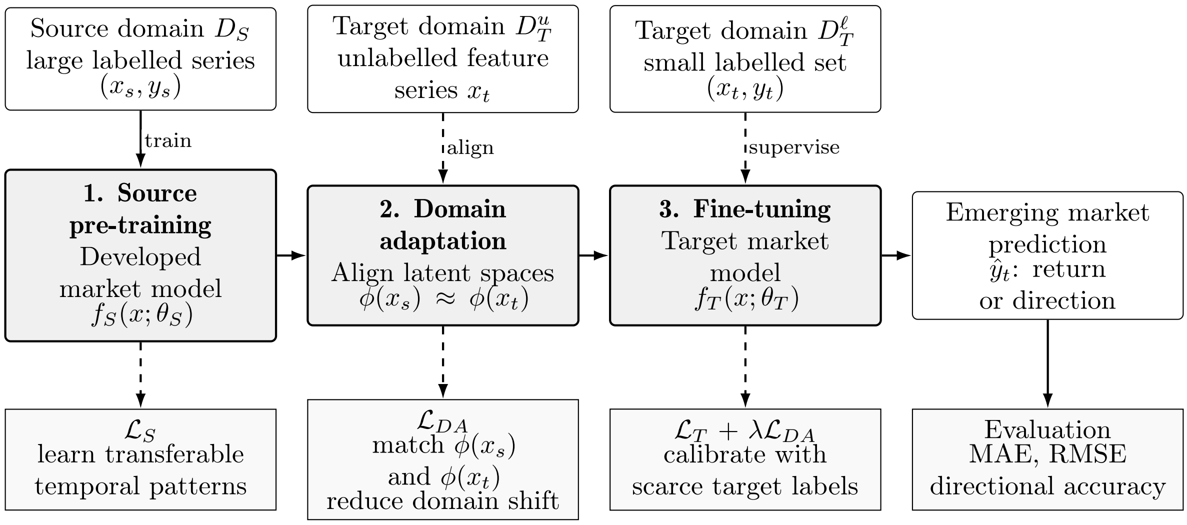
For each forecast origin t, the models described in the remainder of the chapter generate a forecast $\hat{r}_{t+h}$ using only information that would have been available at time t.

Figure 1 shows a summary of the framework, including the data sources, the three stages of the pipleine, the resulting comparison ladder and the common evaluation module. Each tier is evaluated using the same target test window, forecast horizon and predictor availability.

4.1.1 Source and target domains

The emprical framework seperates the analysis into a source and target-domain. The source-domain contains historical observations from a relatively data-rich, developed

Figure 1
Overview of the source-pre-training, domain-adaptation, fine-tuning pipeline and the resulting five-tier model comparison ladder



commodity market and is represented as

$$D_S = \left\{ \left(x_s^{(i)}, y_s^{(i)} \right) \right\}_{i=1}^{n}, \tag{2}$$

where $x_s^{(i)}$ denotes the vector of predictors available at the relevant forecast origin and $y_s^{(i)}$ is the corresponding realised return.

The target domain is defined in the same way but contains a substantially smaller sample from the data-scarce market of interest:

$$D_T = \left\{ \left(x_t^{(j)}, y_t^{(j)} \right) \right\}_{j=1}^{m}, \quad m \ll n. \tag{3}$$

Transfer learning is motivated precisely by the limited amount of target-domain data relative to the larger information set available in the source domain ($m \ll n$). The specific commodities included in the analysis, together with the choice of source and target markets are established in Chapter 3 according to data availability and economic relevance.

4.1.2 The comparison ladder

The main empirical question is whether models that draw on the source-domain dataset D_S achieve better forecasting performance than otherwise comparable models that rely only on target-domain information, when both are evaluated over the same target sample. To evaluate the this question more granularly the empirical design distinguishes five model tiers according to the information available to each model and the way in which that information is used.

- **Tier A: Naive benchmark.** This tier consists of a simple forecasting rule based only on the target series, such as a zero-return, random-walk or historical-mean forecast. It contains no parameters estimated from the source-domain and provides a basic benchmark that any learned model should be expected to outperform.
- **Tier B: Target-only model.** Models on this tier are also trained only on the target-domain dataset D_T. However, its functional form should be identical, or as closely comparable as possible, to that of the transfer-learning models. Tier B is supposed to separate the effect of the model architecture from the effect of transfer learning. If a tranfer model performs better that Tier A but not Tier B, the improvement cannot be attributed to the use of source-domain information.
- **Tier C: Direct transfer.** The source-trained model $f_S(x; \theta_S)$ is applied directly to target-domain observations without any target-specific parameter updating. This provides a measure of how well knowledge learned in the source market transfers to the target market in its original form. It also serves as the reference point for assessing the additional contribution of fine-tuning and domain adaptation in Tiers D and E.
- **Tier D: Fine-tuned transfer.** The source-trained parameters θ_S are updated through supervised training on D_T, but no separate domain-adaptation mechanism is introduced.
- **Tier E: Domain-adapted transfer.** The source-trained model undergoes an explicit domain-adaptation step before, or jointly with, supervised fine-tuning on D_T.

The resulting tier structure allows the contribution of each component to be examined seperately. Comparisons of Tier D or E with Tier B assess whether the use of source-domain information adds value relative to training on the target-domain alone. The comparison between Tiers D and C isolates the effect of target-domain fine-tuning, while the comparison between Tier E and D captures the additioanl contribution of explicit domain adaptation. These comparisons correspond to the thesis hypotheses and are considered directly in Section 4.1.5. Accordingly, the performance of Tier E is not interpreted in isolation as evidence in favour of transfer learning. The lower tiers are reported such that any potential gains can be assessed agains appropriately demanding benchmarks.

As described in Section 4.5, Tier B is estimated in two forms that differ only in the predictors available to the model. The restricted specification uses the set of features common to both the source and target domains, providing a like-for-like comparison with Tiers C-E. The full specification additionally includes predictors that are available only in the target market. Comparing these two versions helps determine whether differences in forecasting performance arise from transferred information or from the availability of additional local predictors.

4.1.3 The three-stage pipeline

Tiers C, D and E are constructed using a shared three-stage pipline, drawing on the general source pre-training, domain-adaptation and fine-tuning framework by Mashetty et al. [20] for cross-market financial prediction.

- **Stage A: Source pre-training.** A model $f_S(x;\theta_S)$ is trained on D_S to minimise a supervised prediction loss, producing pre-trained parameters θ_S that are intended to capture return-generating patterns that are not specific to source marekt alone (e.g., momentum, mean reversion or responses to common global factors). Applying $f_s(\cdot;\theta_S)$ directly to D_T without further modification yields Tier C.
- **Stage B: Domain adaptation.** A seperate alignment procedure is introduced to reduce differences between the source and target feature distributions. This addresses the possibility that a model trained solely to minimise prediction error in the source domain may learn representations that do not transfer effectively to the target domain. The precise alignment mechanism, such as a discrepancy-based penalty using maximum mean discrepancy (MMD) or an adversarial approach, is specified in Section 4.3 after the model class has been established.
- **Stage C: Fine-tuning.** The pre-trained, and where applicable adapted, model parameters are subsequently updated through supervised training on D_T. Fine-tuning the output of Stage A directly produces Tier D, whereas fine-tuning the model after Stage B produces Tier E.

The framework as it is used in this thesis differs from Mashetty et al. [20] in several ways due to its application to commodity rather than equity markets. First, clearly the source and target domains are defined using commodity return series rather than equity-index ones. This distinction is important because commodity-market linkages may operate through mechanisms such as global price discovery, storage and convenience-yield dynamics or commodity-specific supply shocks, which differ from the forces underlying equity-markets comovement. Second, the empirical design considers multiple target markets and, where data availability allows, multiple commodity classes. Third, the contribution of each stage of the pipeline is assessed separately through the explicit ablation structure represented by Tiers C, D and E, rather than evaluating only the fully combined model. These distinctions are examined further in Section 4.1.5 and Chapter 2.

4.1.4 Information set and temporal discipline

Since the aim of the analysis is to perform a genuine out-of-sample forecast, the same chronological restrictions are applied throughout the pipeline. For any forecast at time t, models can only use predictor information available at or before time t. Target domain observations occurring after the forecast origin are excluded from all stages of model development and evaluation.

In Stage A, source-domain pre-training is conducted using only observations available before the beginning of the target evaluation period. This ensures that the pre-trained model is based solely on information that would have been available when the target forecasting started. If an expanding- or rolling-window retraining procedure is used rather than a single fixed pre-training step, the relevant procedure is described in Section 4.2.

Predictor transformations, such as normalisation, scaling, and feature construction, are estimated using the training data only. The fitted transformations are then applied unchanged to the validation and test sets. These chronological and preprocessing rules apply consistently across all stages and tiers, and are therefore stated here rather than repeated in each subsequent section unless a particular stage requires additional clarification.

4.1.5 Mapping to the research hypotheses

Table 2 shows how each of the thesis hypotheses corresponds to a specific comparison analyzed in Chapter 5.

4.2 Stage A: Source Model Pre-Training

This section defines Stage A of the pipeline introduced in Section 4.1.3, in which the source-domain model $f_S(x; \theta_S)$ is estimated using D_S. The resulting parameters θ_S are subsequently carried forward into Tiers C, D and E. Therefore, the modelling decisions

Table 2
Mapping the comparison ladder to the research hypotheses

Hypothesis	Comparison	Evaluated in
H1: transfer improves on target-only training	Tier D and/or Tier E vs. Tier B (restricted)	Sections 5.1, 5.2
H2: transfer's advantage is larger when target data are scarcer	Gap between the best-performing transfer tier and Tier B, evaluated across shrinking target training histories	Section 5.3
H3: adaptation improves on naive transfer	Tier D and/or Tier E vs. Tier C	Section 5.2
H4: transfer performance depends on source-target similarity	Ranking of Tiers C-E under alternative source-market choices, related to ex-ante economic and statistical similarity measures	Section 5.5.2

made at this stage determine the information and structure available to all three transfer-learning tiers.

4.2.1 Model class and parameter budget

The underlying model $f(\cdot;\theta)$ is specified as a supervised sequence-learning architecture. Two candidate models are used: a LSTM network and a temporal convolutional network (TCN)[1]. Both are compatible with the pipeline described in Section 4.1.3 and with the tiered comparison framework introduced in Section 4.1.2. The final choice between the two architectures or the decision to report results for both as part of the robustness analysis in Section 5.5 is made after the source- and target domain sample sizes have been established in Chapter 3.

The comparison design imposes an additional constraint on model selection that would not necessarily arise in a conventional pre-training and fine-tuning setting. Tier B, described in Section 4.3, is trained from scratch using only target-domain data and employs the same or a closely comparable model architecture. The underlying model can therefore not be selected solely on the basis of its performance on D_S. An architecture that fits the relatively large source-domain sample well but has to many parameters for the smaller target-domain dataset D_T could place Tier B at a structural disadvantage. In that case, superior performance by Tiers D or E might reflect differences in effective model capacity rather than the additional value of information from the transfer.

For this reason, the size of the architecture is constrained by the amount of data available to the restrictred Tier B specification. Parameters such as hidden-state dimension, network

[1] Finalize which models will be used.

depth and lookback length are chosen such that specification can be estimated without pronounced overfitting. This condition is assessed empirically through learning curves that relate out-of-sample forecasting error to the size of the target–domain training sample for a given level of model capacity, as part of the hyperparameter-selection procedure described in Section 4.7. Where this requirement substantially restricts the capacity that could otherwise be achieved in Stage A, the resulting trade-off is reported explicitly.

4.2.2 Input representation

The input to f_S is based exclusively on a shared-core feature set which are consistent with those found in the literature[2]. This includes own-series technical indicators, such as lagged log returns, rolling realised volatility, moving averages, the relative strength index and Bollinger-band measures, together with global and macro-financial variables that are plausibly relevant across commodity marekts. Examples include a global commodity index or benchmark return, the US dollar index, a global risk-sentiment measure and global short-term interest rates. No target-specific local predictors are included at this stage. As discussed in Section 4.2.6 and formalised in Section 4.5, such variables are not available in the source domain and therefore cannot be incorporated into a source-trained model that is intended to be applied directly to D_T.

Let $x_\tau \in \mathbb{R}^d$ denote the shared-core feature vector observed at time τ, where the feature dimension d is the same in the source and target domains. Each source-domain training observation is constructed using a lookback window of length L.

$$X_s^{(i)} = (x_{i-L+1}, x_{i-L+2}, \ldots, x_i)\,, \quad y_s^{(i)} = r_{i+h}, \tag{4}$$

The source model therefore maps an $L \times d$ sequence of predictors to a scalar return forecast. The lookback length L is treated as a hyperparameter and is selected using source-domain validation data only, as desribed in Section 4.2.5. Its final value is reported in Section 4.7 alongside other architectural choices.

Restricting x_τ to the shared-core feature set ensures that the same input construction can be used for the target-domain observations in Tiers C, D and E. In this way, the predictor representation remains identical across domains, so differences in forecasting performance can be attributed to the modelling stage being evaluated and not to the information supplied to each model.

[2] Insert specific citations.

4.2.3 Normalisation protocol

4.2.4 Prediction target and task loss

$$\mathcal{L}_S\left(\theta_S\right) = \frac{1}{n}\sum_{i=1}^{n}\left(f_S\left(X_s^{(i)};\theta_S\right) - y_s^{(i)}\right)^2, \tag{5}$$

4.2.5 Regularisation and validation during pre-training

4.2.6 What Stage A is intended to capture

The extent to which the pre-trained parameters θ_S are valuable in the target domain depends on whether they capture relationships that generalise beyond the source market. In this setting, two broad groups of shared inputs are especially relevant, although their potential for transfer is likely to differ.

The first group comprises patterns associated with the underlying return-generating process. These include short-term momentum or mean reversion, volatility clustering and the sensitivity of commodity returns to global risk and financing conditions, such as movements in the US dollar index, global short-term interest rates and measures of global risk sentiment. Such relationships are plausible candidates for cross-market transfer because the mechanisms underlying them, including global commodity price discovery, dollar-denominated pricing, and common exposure to global business-cycle conditions, extend beyond any individual source market.

The second group consists of relationships that may arise from features specific to the source market itself. Examples include trading-session effects, liquidity conditions on the source exchange and market responses to source-country economic announcements. These patterns are less likely to generalise to the target domain. Their influence on θ_S is supposed to be reduced by the domain-adaptation procedure in Stage B as described in Section 4.3. However, this distinction should be interpreted as a working hypotheses about the types of information that source pre-training may capture and not as an empirical finding established in advance. The issue is examined again in Section 5.4 through the cross-commodity comparison and in Chapter 6 in the discussion of negative transfer.

4.3 Stage B: Domain Adaptation

4.3.1 Motivation for Distribution Alignment

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List of Acronyms

ARIMA Autoregressive Integrated Moving Average

BiLSTM Bidirectional Long Short-Term Memory

CNN Convolutional Neural Network

ESN Echo State Model

GARCH Generalized Autoregressive Conditional Heteroskedasticity

GRU Gated Recurrent Unit

LSTM Long Short-Term Memory

MAE Mean Absolute Error

MLP Multilayer Perceptron

MMD Maximum Mean Discrepancy

NN Neural Network

RMSE Root Mean Squared Error

RNN Recurrent Neural Network

SVM Support Vector Machine

SVR Support Vector Regression

TCN Temporal Convolutional Network

XGBoost Extreme Gradient Boosting